spaCy Dependency Parse

This program uses **spaCy**, a Natural Language Processing (NLP) library, to perform **dependency parsing**. Dependency parsing identifies grammatical relationships between words in a sentence.

Step 1: Install spaCy

Open Command Prompt (Windows) or Terminal (Linux/Mac):

```
pip install spacy
```

Step 2: Download the English Language Model

The code uses the model en_core_web_sm.

Run:

```
python -m spacy download en_core_web_sm
```

Step 3: Create the Python Program

Save the following as dependency_parse.py:

```
import spacy

nlp = spacy.load("en_core_web_sm")

sentences = [
    "Visiting aunts can be a nuisance.",
    "She went to the bank.",
    "I saw the man with the telescope.",
    "Can you pass the salt?"
]

for s in sentences:
    print("Sentence:", s)
    print("-" * 40)

    for tok in nlp(s):
        print(tok.text, tok.dep_, tok.head.text)

```

```
19      print()
```

Step 4: Run the Program

```
1  python dependency_parse.py
```

Example Output

For

```
1  I saw the man with the telescope.
```

you may get something similar to:

```
1  I           nsubj    saw
2  saw         ROOT     saw
3  the         det      man
4  man         dobj     saw
5  with        prep     man
6  the         det      telescope
7  telescope   pobj     with
8  .           punct    saw
```

Meaning

Word	Dependency	Head
I	nsubj	saw
man	dobj	saw
with	prep	man
telescope	pobj	with

This means spaCy interprets:

```
1  [I saw] [the man with the telescope]
```

In other words, **the man has the telescope**.

Exercise Answer

What is the head of "with"?

From the dependency output:

```
1  with    prep    man
```

The head of **with** is:

```
1  man
```

Why is the Sentence Ambiguous?

Sentence:

```
I saw the man with the telescope.
```

has two possible meanings.

Interpretation 1 (spaCy's default)

```
I saw [the man with the telescope]
```

Tree:

```
saw
├── I
└── man
    ├── the
    └── with
        └── telescope
```

Meaning:

> The man possesses the telescope.

Interpretation 2

```
I used a telescope to see the man.
```

Tree:

```
saw
├── I
├── man
│   └── the
└── with
    └── telescope
```

Meaning:

> "with the telescope" modifies **saw** rather than **man**.

This is the "other reading" that the exercise asks you to draw.

Visualizing the Dependency Tree

Install:

```
pip install spacy
python -m spacy download en_core_web_sm
```

Then use:

```
from spacy import displacy
import spacy

nlp = spacy.load("en_core_web_sm")

doc = nlp("I saw the man with the telescope.")

displacy.serve(doc, style="dep")
```

Open:

```
http://localhost:5000
```

in your browser to see the dependency tree graphically.

Key Learning Outcome

- **AST Parsing** (Task 1) parses programming languages according to fixed grammar rules.
- **Dependency Parsing** (Task 2) parses natural language and determines grammatical relationships.
- Natural language can be **ambiguous**, so multiple valid parse trees may exist for the same sentence.
- In spaCy's default parse, the head of **"with"** is **"man"**.